First year Aerospace Engineering

How aircraft fly

Marcos Carrera Valle

Pioneers of flight

George Caley; Wilbur & Orville Wright, Samuel Langley, Anthony Gokker; Albert Plesman

19^{th} Century - unpowered

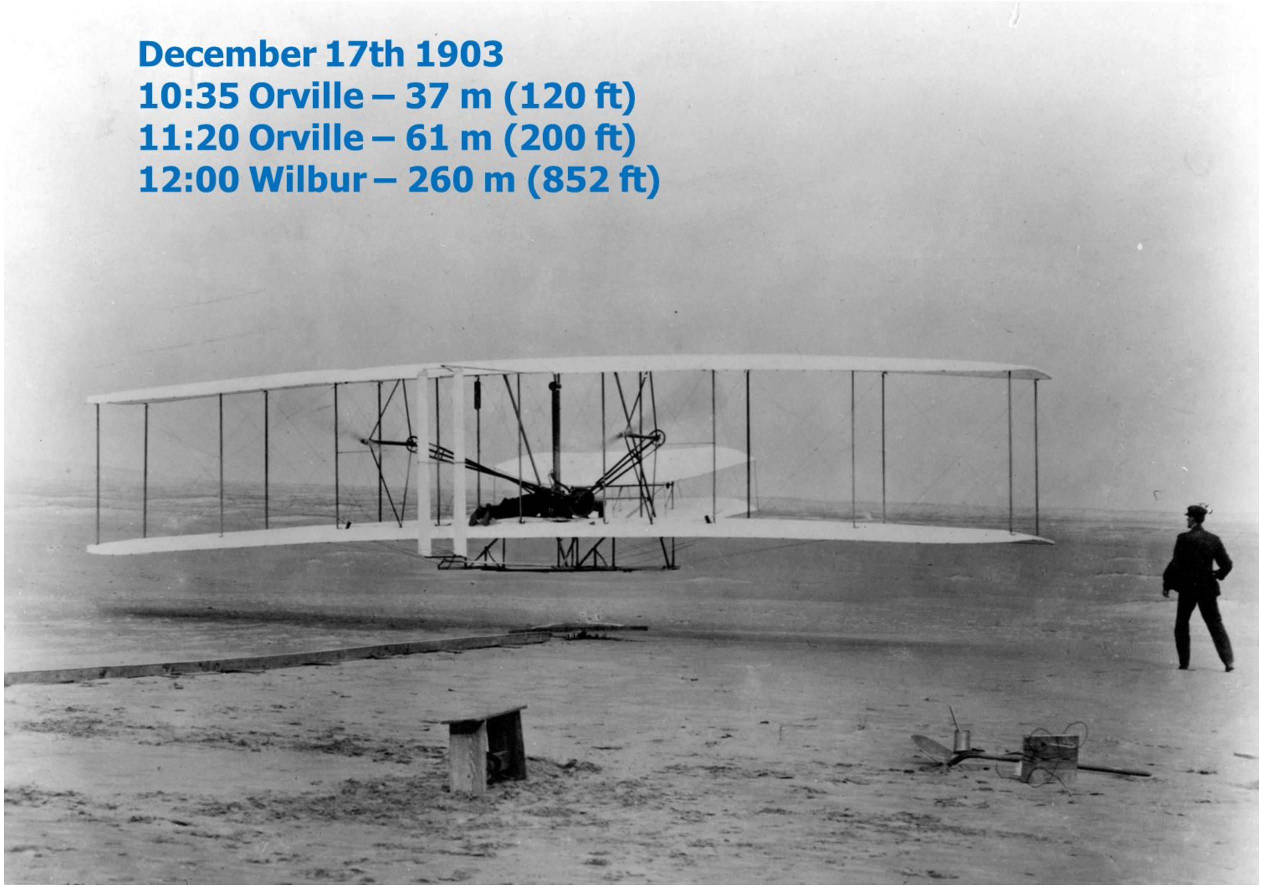
Otto Lilienthal (1848 – 1896)

Was fascinated with the flight of birds (Storks)
Studied at Technical School in Potsdam
Started experiment in 1867
Made more than 2000 flights
Build more than a dozen gliders
Build his own “hill” in Berlin
Largest distance 250 meters
Died after a crash in 1896.

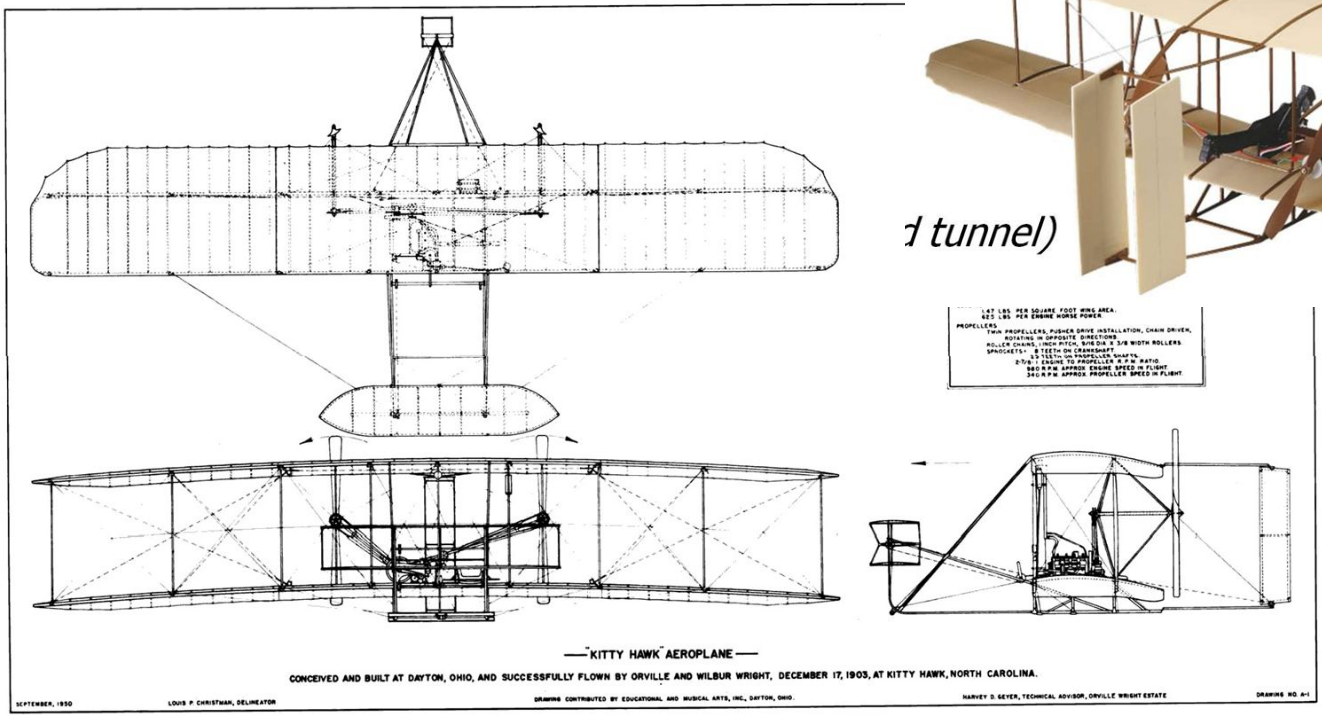
No filmed evidence:
film invented in 1895 (Lumiere)

Otto Lilienthal

Few designs


December 17th 1903
10:35 Orville – 37 m (120 ft)
11:20 Orville – 61 m (200 ft)
12:00 Wilbur – 260 m (852 ft)

Wright flyer



d tunnel)

Wright Brs. – the Flyer (1903)

Printing firm & Bicycle shop
Interested in flying
(inspired by Lilienthal' gliders)
Did a lot of investigations:
- Build a wind tunnel to test planes

Timing (readiness of technology)
Pressure of Samuel Pierpont Langley

- The engine 12 Horse Power

- The propeller (2 counter rotating push propellers)
- Perfected in wind tunnel

From this…

to this in 102 years…

Straight, horizontal, steady flight

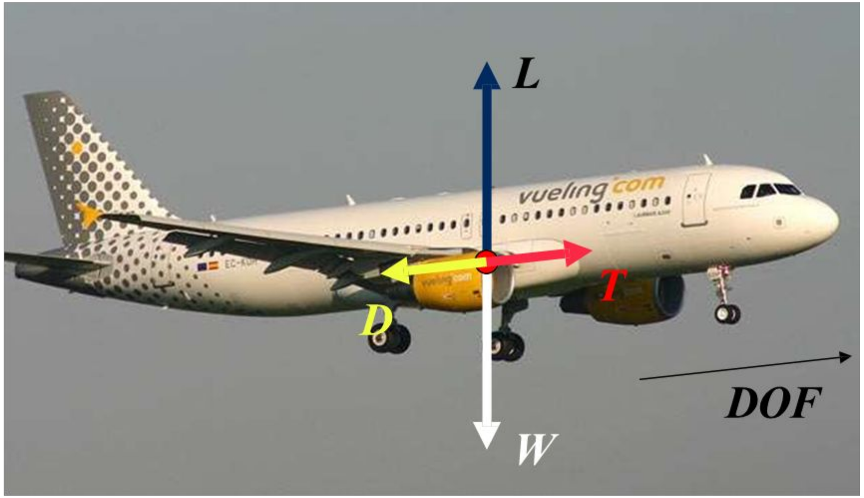
Four forces:

- Weight W
- Lift L
- Drag D
- Thrust T

If V , h = constant

Force Equilibrium:

$$L = W$$
$$D = T$$



Four forces

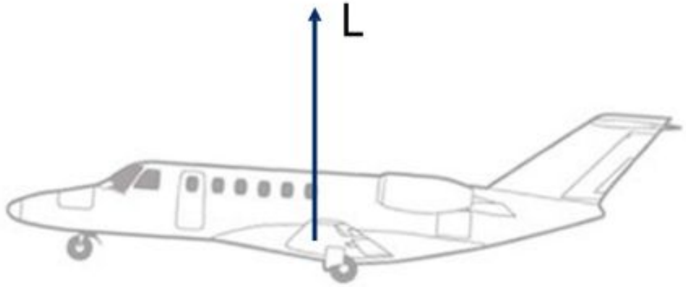
- **Lift**: mainly generated by the **wing** (smaller contributions by fuselage and tail surfaces)

- **Drag** is caused by **fuselage**, **wings**, **tail surfaces**, etc.

- **Weight** is composed of three main components:
  - **aircraft empty** weight
  - **fuel**
  - **Payload** (Passangers + luggage + freight)
  - **Thrust** is provided by the **engines**

Lift

Lift depends on: .



p = Density of the air [kg/m^3]

V = Air speed [m/s]

S = Wing area [m^2]

C_L = Lift coefficient

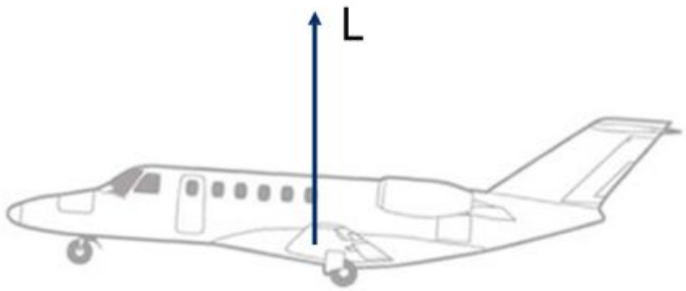
This is an **aerodynamic** coefficient, it depends on:

- shape of aircraft and wing
- angle between airspeed vector and aircraft

(- scale effects: vicosity is different at different scales)

(- Mach effects: pressure waves when speed is around or higher than sound)

Density



The air is made up of particles

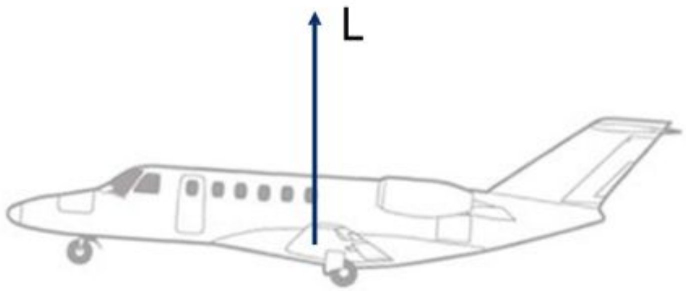
Part of what makes aircraft fly is the particles moving upwards (due to pressure differences), and “pushing” the wings up

The more particles “pushing up”, the more the wings are pushed up

If the air has many particles, the density is higher

So a higher density means higher lift

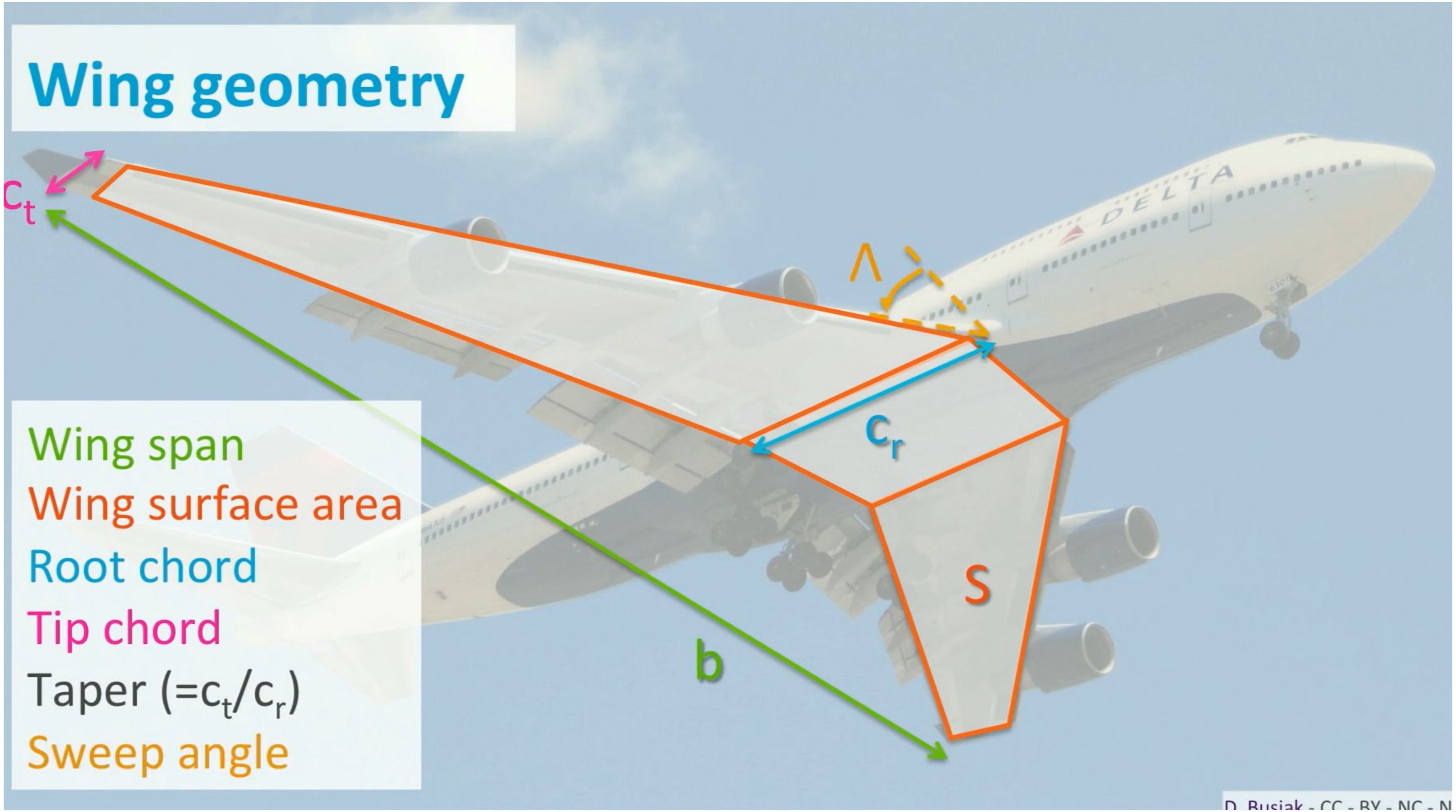
Wing area



The particles push up on the surface of the wing

The larger the surface of the wing, the more places where the particles can push upwards

So a larger wing surface area leads to a larger lift


Wing geometry
Wing span
Wing surface area
Root chord
Tip chord
Taper (=c_t/c_r)
Sweep angle
c_t
Λ
c_r
S
b
D. Rusiak - CC - BY - NC - N

Generation of lift

Why do the particles “move” upwards

This is due to a pressure difference

Pressure on the top of the wing is lower than below the wing

Generation of lift

Many explanations as to why this happens:

- Longer way
- Air pushed down
- Circular lift theory

My favourite: Continuity and Bernoulli

Continuity

Mass coming in = Mass coming out

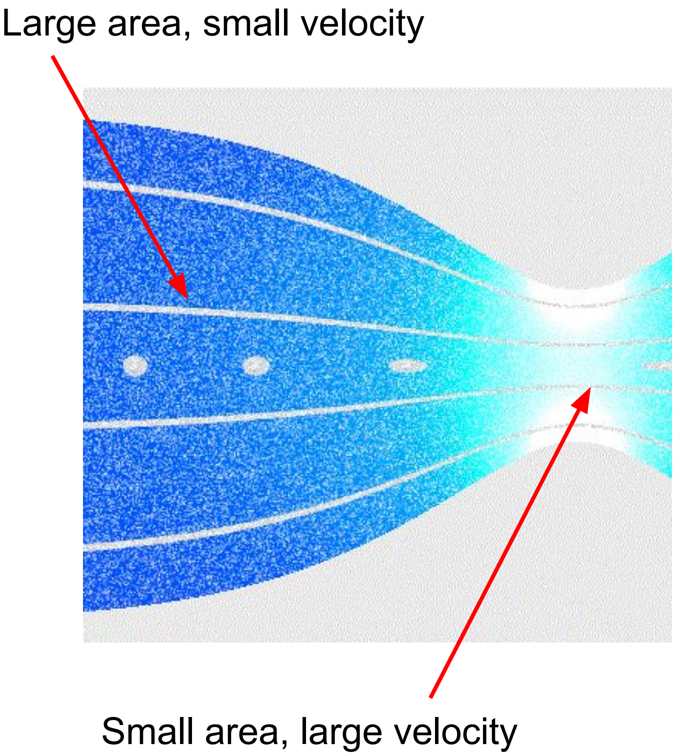
Mass flow = mass * velocity = **CONSTANT**

Mass = density * area

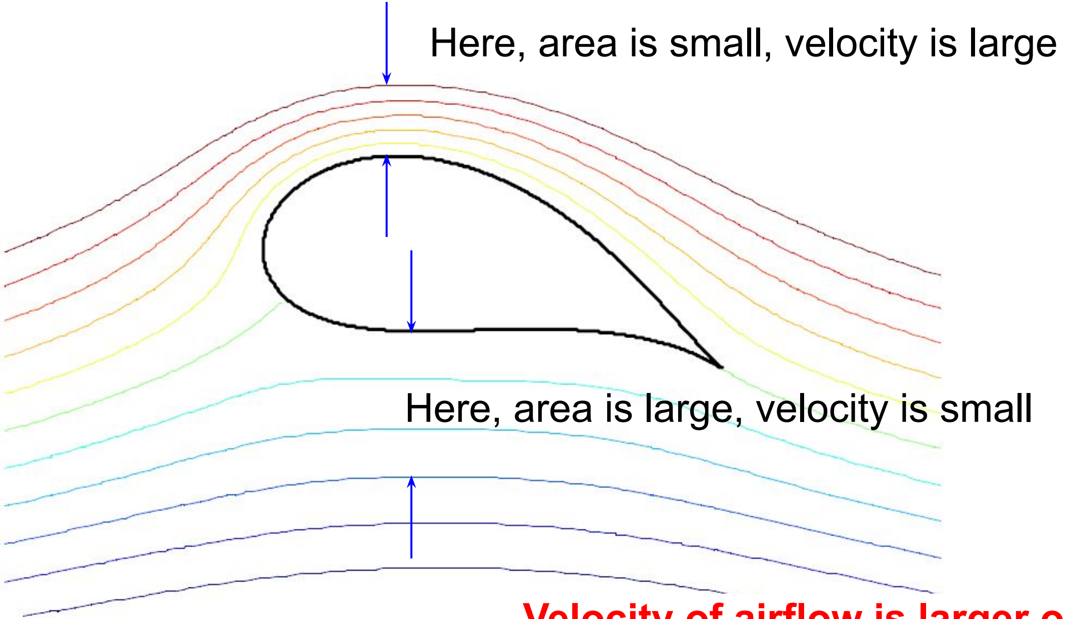
So **rho*A*V = Constant**

When the area ‘A’ is decreased, the velocity of the fluid ‘V’ increases

Assuming incompressible flow
(density = constant)



For a wing



Velocity of airflow is larger on top of the wing than below the wing

Bernoulli's principle (1738)

Johann Bernoulli
1667 - 1748

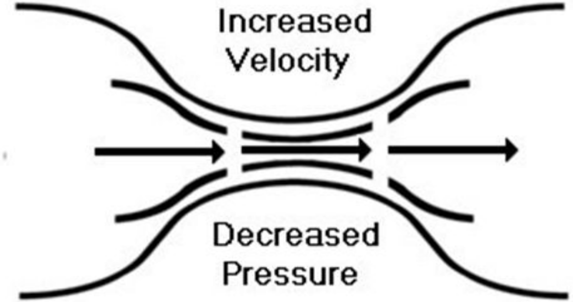
$$p_1 + \frac{1}{2} p V_1^{\,2} = p_2 + \frac{1}{2} p V_2^{\,2}$$

$$p_1 + \frac{1}{2} p V_1^{\,2} + pgh_1 = p_2 + \frac{1}{2} p V_2^{\,2} + pgh_2$$

$$p + \frac{1}{2} p V^2 = \text{constant along streamline}$$

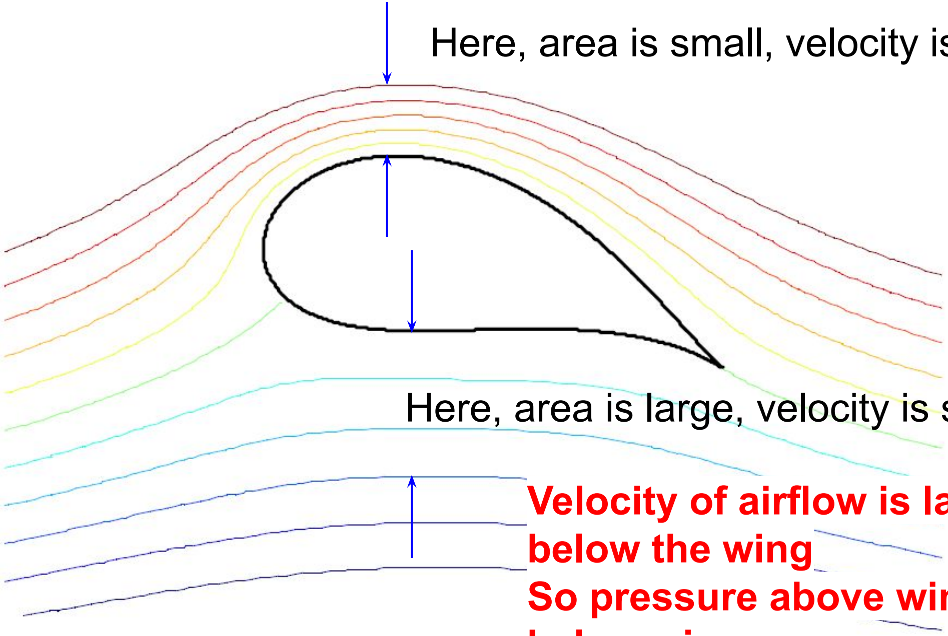
$$p_{static} + p_{dynamic} = p_{total} = \text{constant along streamline}$$

$$p_{static} + q = \text{constant along streamline}$$



Only true for constant ρ , so constant density,
incompressible => low speeds, complete flow subsonic

For a wing

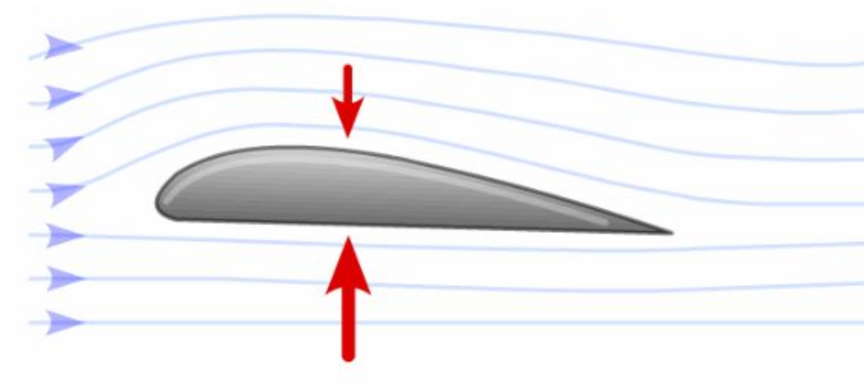


Here, area is small, velocity is large, pressure is small

Here, area is large, velocity is small, pressure is large

Velocity of airflow is larger on top of the wing than below the wing
So pressure above wing is smaller than pressure below wing

Velocity



Upward pressure is larger than downward pressure

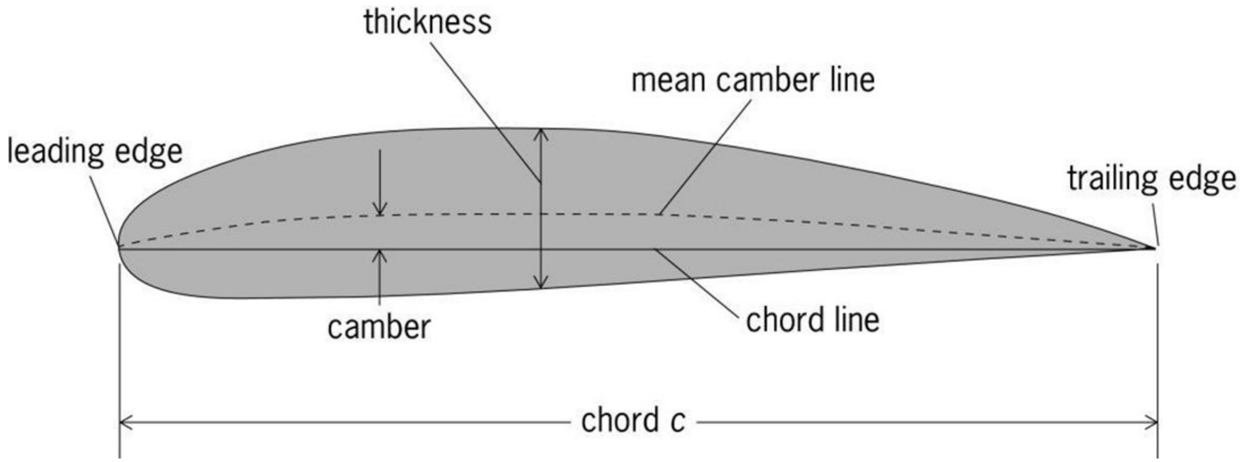
Resulting force pushes upwards → Lift!

This effect is more clear as velocity increases

So increasing velocity increases lift

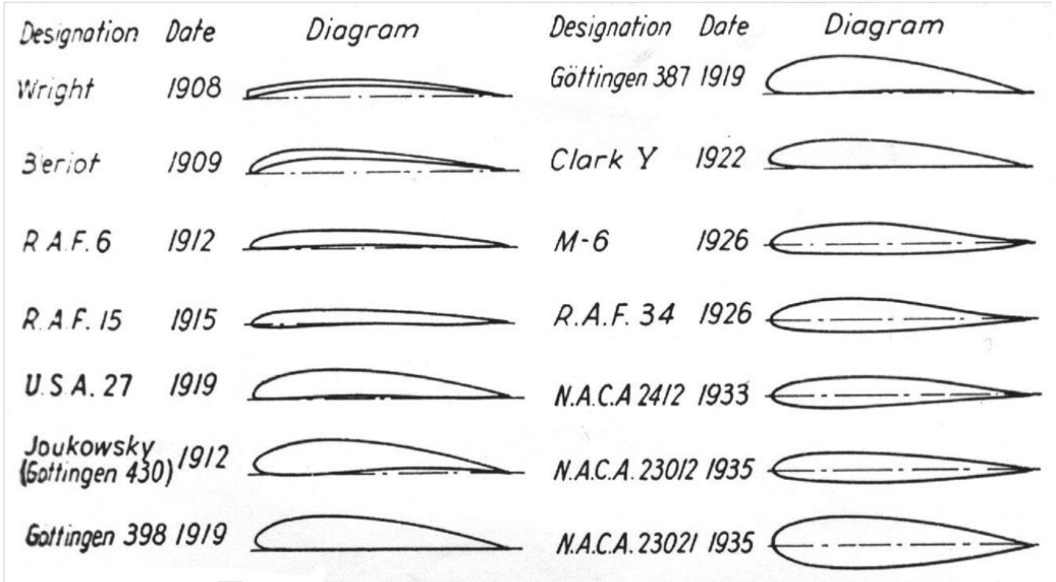
Lift coefficient c_l

Defined by the shape of the **airfoil**



History of wing airfoils ('vleugelprofielen')

Early years: many different airfoil descriptions

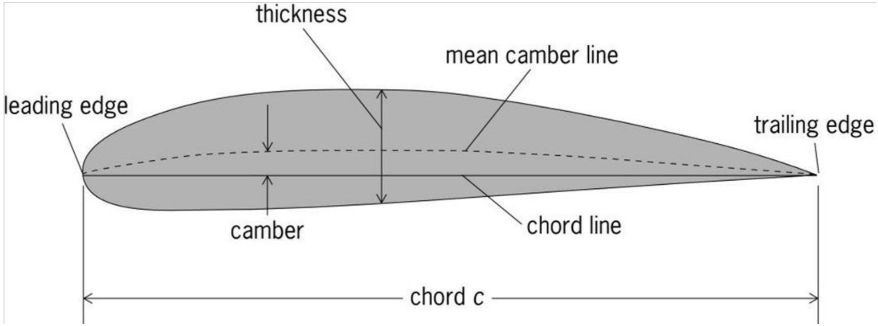


Lift coefficient c_l

Until National Advisory Committee for Aeronautics (NACA) established a way to call their airfoils

Example: NACA 2412

2412 = **2 4 12** and means :
2% camber (of chord line) at
0.**4** of the chord (from LE); and
12% thickness/chord ratio (or 0.12)



http://airfoiltools.com/airfoil/naca4digit

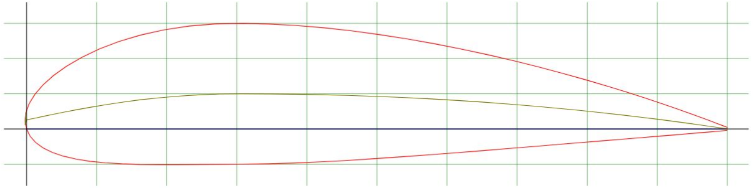
What about NACA 23014 ???

- **Five-digit series: 2 30 14**
- The NACA five-digit series describes more complex airfoil shapes:[7]
- The first digit, when multiplied by 0.15, gives the designed coefficient of lift (C_L).
- Second and third digits, when divided by 2, give , the location of maximum camber as a distance from the leading edge (as per cent of chord).
- Fourth and fifth digits give the maximum thickness of the airfoil (as per cent of the chord)

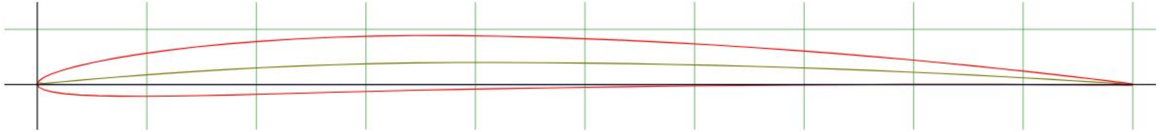
- For example, the NACA 23012 airfoil would give an airfoil with:
  - 2: a design lift coefficient C_L=0.3 (2 x 0.15)
  - 30: maximum camber located at (30/2=) 15% chord
  - 14: maximum thickness of 14% chord

Brightspace contains:
- NACA document with airfoil data
- List of a/c and their airfoils (URL)

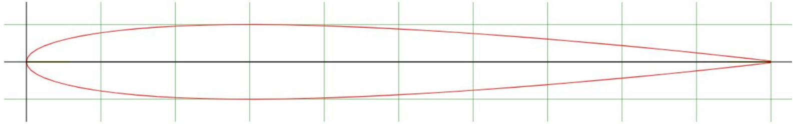
Lift coefficient c_l - quick exercise



NACA 2405



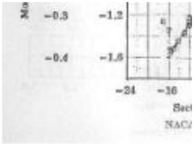
NACA 0010



NACA 5320

Lift coefficient c_l

$$Re = \frac{Vc}{\nu} = \frac{\rho V c}{\mu}$$

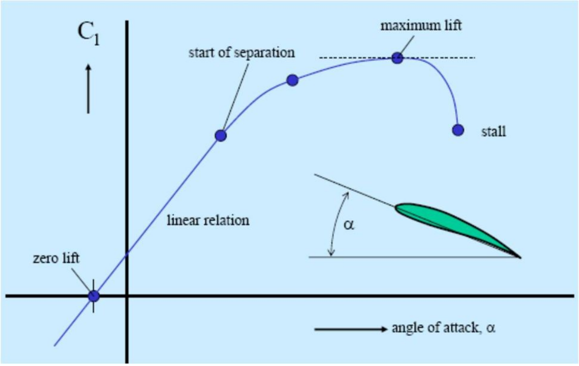


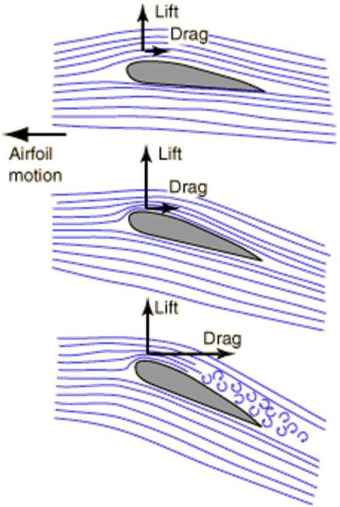
$$\mu = viscosity = 1.7894 \cdot 10^{-5} \quad kg/ms$$

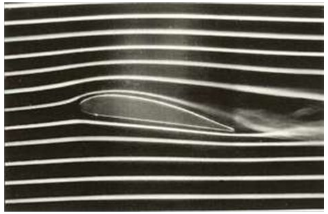
Also depends on other parameters:

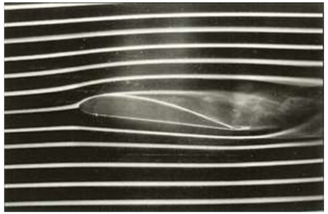
- Reynolds Number (Viscosity)
- Angle of attack

c_l Taken from test data (wind tunnels)



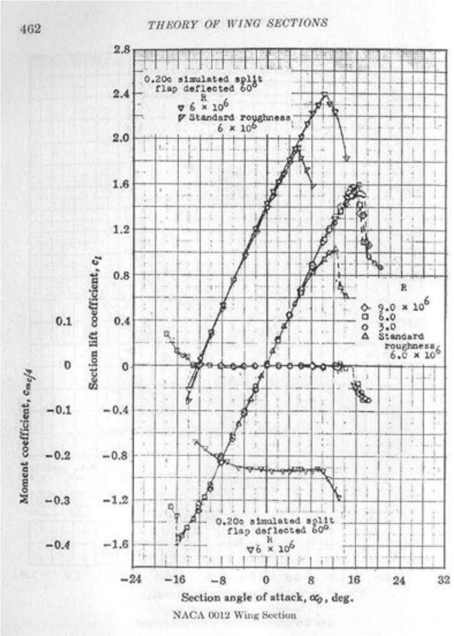


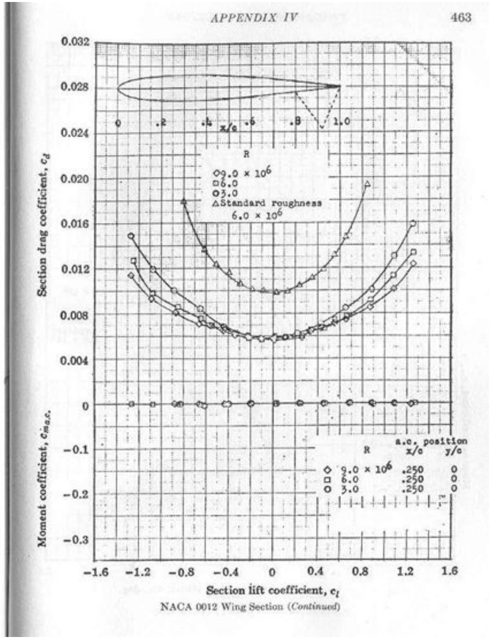




various angles of attack (α) and h/c ratio

α°	h/c	C_l	C_d	C_l/C_d
0	0.2	0.67112	0.012566	53.41
0	0.5	0.65475	0.012614	51.66
0	1.0	0.65437	0.013279	49.28
4	0.2	1.1841	0.015862	74.65
4	0.5	1.0658	0.019398	54.94
4	1.0	1.0397	0.020825	49.93
8	0.2	1.5217	0.031304	48.61
8	0.5	1.3245	0.034725	38.14
8	1.0	1.3183	0.034992	37.67





Lift coefficient c_l

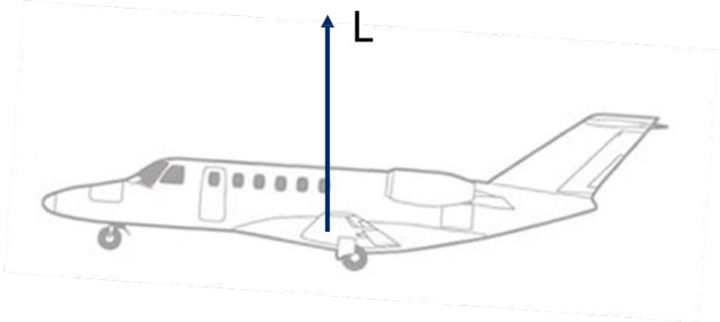
Takeaway:

The larger the lift coefficient, the larger the lift

Very intuitive tbf

Lift

The formula for the LIFT is:

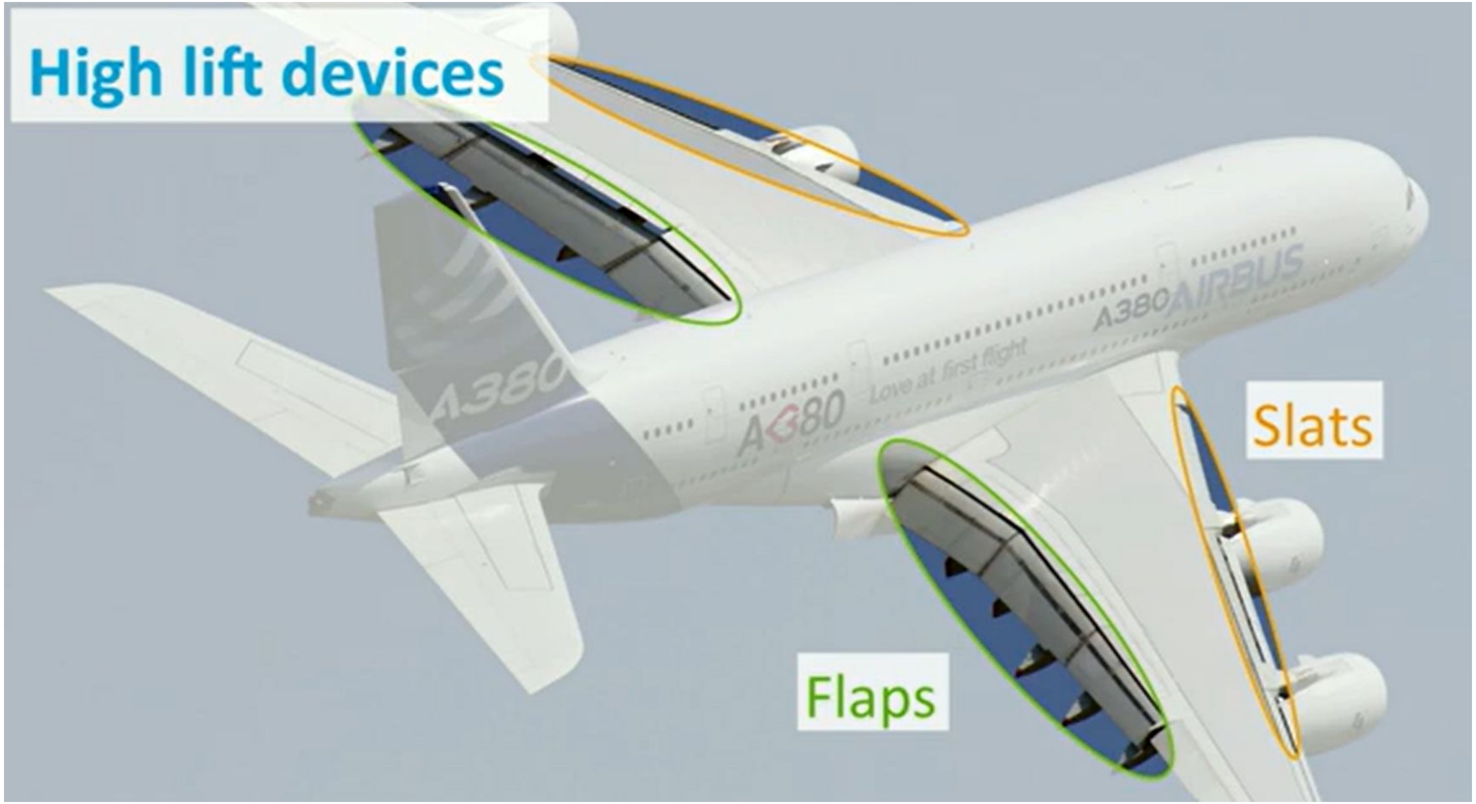


$$\boxed{L = C_L \frac{1}{2} pV^2 S}$$

p = Density of the air [kg/m³]
V = Air speed [m/s]
S = Wing area [m²]

C_L = Lift coefficient

So what do we do when velocity is low? (take-off and landing)



Change the geometry of the wing (Higher Surface Area)
Change the shape of the airfoil (To increase c_L)

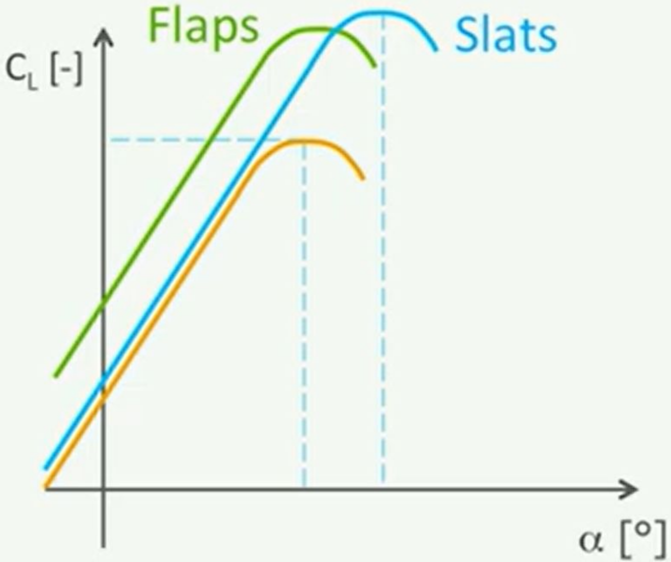
High lift devices

Purpose: Fly at low speeds

By: Increasing critical α

Increasing $C_{L_{max}}$

Increasing wing area S

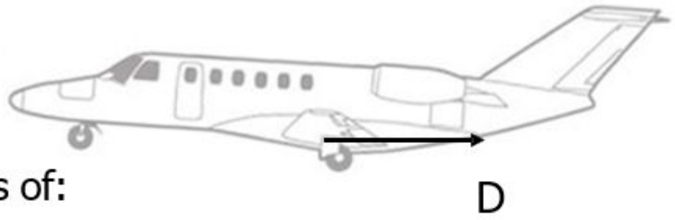


Slats:

Flaps:

Drag

$$D = C_D \frac{1}{2} \rho V^2 S$$



C_D (Drag coefficient) consists of:

Zero lift drag:

- profile drag of wing
- pressure and friction of tail, nacelle, fuselage

Induced drag:

- due to lift being generated

ρ (air density) depends on:

- altitude (see previous lecture on atmosphere)

V (air speed) and S (wing area) are design parameters

‘Base’ Drag coefficient $c_{D,0}$

Drag coefficient: $C_D = C_{D_0} + C_{D_i}$

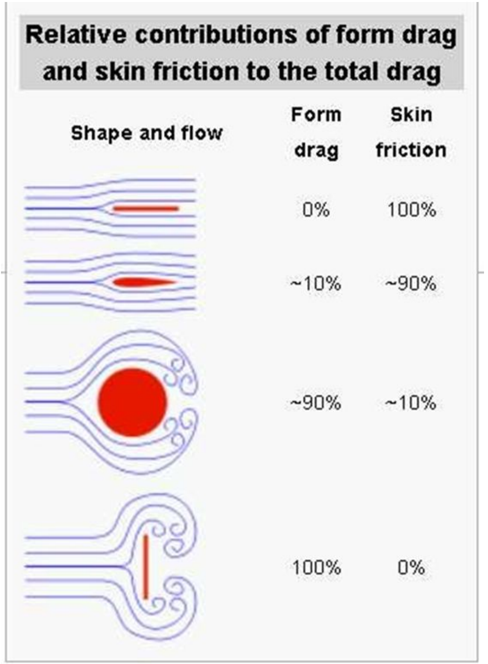
C_{D0} depends on:

Skin friction - normal friction

Pressure drag - from changing the airflow around wing

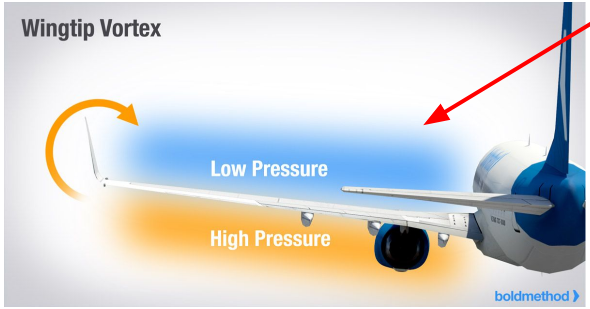
Wave drag - when you go supersonic

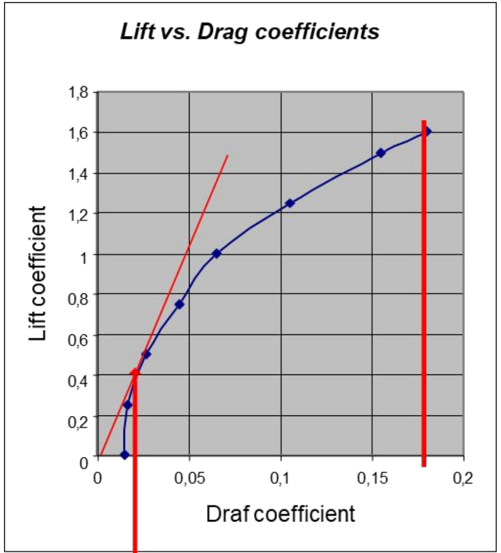
Parasitic drag - from fuselage and engines



Drag coefficient c_d

Drag coefficient: $C_D = C_{D_0} + C_{D_i}$





Maximum C_L/C_D - ratio

Drag polar: $C_D = C_{D_0} + kC_L^2 = \boxed{C_{D_0} + \dfrac{C_L^2}{\pi A e}}$

Lift to Drag Ratio

L/D ratio = c_l/c_d ratio

Good way to see how efficient an airfoil/ wing is
At the maximum c_l/c_d ratio, fuel consumption is the smallest
Typical values:

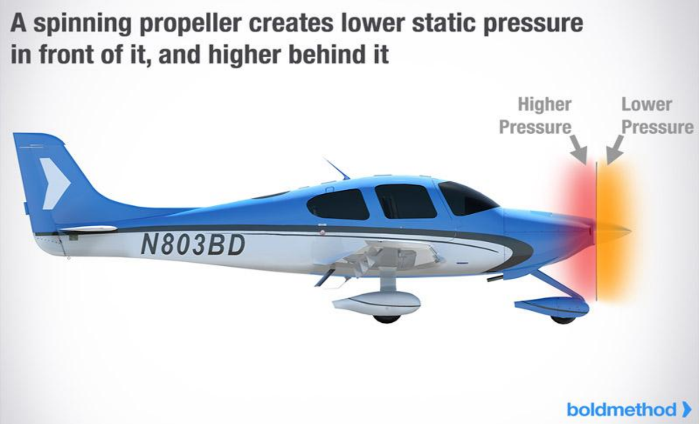
• Modern sailplane	~60
• Lockheed U2	28
• B-747	17
• Concorde (cruise)	7.1
• Cessna	7
• Space shuttle (hypersonic flight)	1

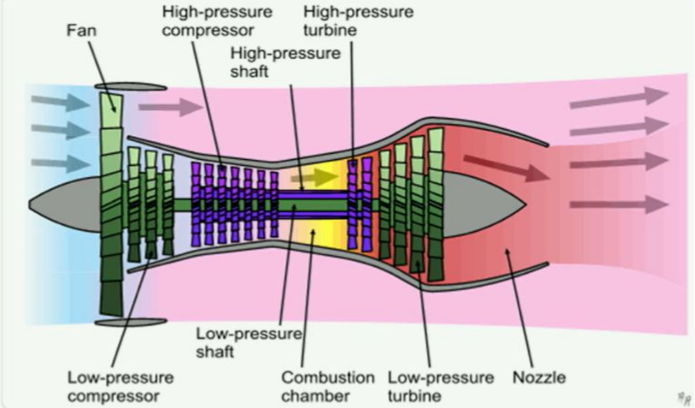
Thrust

We obviously want to move forward!
We do that using engines
Different types of engines for different types of missions

Engine types

- **Propeller engines**
  - Piston engine
  - Turboprop
- **Jet engines**
  - Turbojet
  - **Turbofan**
  - Ramjet





Thrust

We obviously want to move forward!
We do that using engines
Different types of engines for different types of missions

To fly slow and low → Propeller engines
To fly fast and high → Jet engines
To fly fast and low → Jet engines

The snowball effect

Reducing some weight means:

1) Less lift required
2) So smaller wings
3) So less drag
4) So smaller engines

Smaller engines and wings means less weight, so …

2) Exercises

An aircraft is flying at cruise level (density = 0.3 kg/m^3)
Use g = 9.81
The wing is has a taper ratio of 1 and a wingspan b of 50 metres
The chord length of the wing is 6 metres
The mass of the aircraft is 70,000 kg
The wing has a lift coefficient of 0.85

a) What is the aspect ratio of the wing?
b) At what velocity is the aircraft flying at?

The aircraft now reduces its speed to 70% during landing.

c) Assuming the wing area stays constant, what is the new required lift coefficient? How does it achieve this?

3) Exercises

5.21 The Cessna Cardinal, a single-engine light plane, has a wing with an area of $16.2\ m^2$ and an aspect ratio of 7.31. Assume that the span efficiency factor is 0.62. If the airplane is flying at standard sea-level conditions with a velocity of 251 km/h, what is the induced drag when the total weight is 9800 N?

5.22 For the Cessna Cardinal in Prob. 5.21, calculate the induced drag when the velocity is 85.5 km/h (stalling speed at sea level with flaps down).

4) Exercises

An aircraft is flying at cruise level (density = 0.2546 kg/m^3)
Use g = 9.81
The wing is has a taper ratio of 1 and a wingspan b of 30 metres
The chord length of the wing is 4 metres
The mass of the aircraft is 70,000 kg
The wing has a lift coefficient of 0.85

4) Exercises

a) At what speed is the aircraft flying at? (m/s)

b) The aircraft now approaches landing at sea level, and reduces its velocity to 90 m/s. What is the lift coefficient during landing?

$density_{cruise} = 0.2546\ kg/m^3$

$g = 9.81\ m/s^2$

Taper ratio = 1

$b = 30\ m$

$c = 4\ m$

$m = 70{,}000\ kg$

$c_L = 0.85$

$V = 229.97\ m/s$

$density_{land} = 1.225\ kg/m^3$

4) Exercises

a) At what speed is the aircraft flying at? (m/s)
b) The aircraft now approaches landing at sea level, and reduces its velocity to 90 m/s. What is the lift coefficient during landing?
c) How could the aircraft achieve this increase in c_L

$\text{density}_{cruise} = 0.2546 \text{ kg/m}^3$
$g = 9.81 \text{ m/s}^2$
Taper ratio = 1
$b = 30$ m
$c = 4$ m
$m = 70{,}000$ kg
$c_L = 0.85$
$V = 229.97$ m/s
$\text{density}_{land} = 1.225 \text{ kg/m}^3$
$c_{L,\ land} = 1.15$

4) Exercises

The aircraft has an oswald efficiency factor of 0.92
The parasitic drag coefficient is 0.00685

density_{cruise} = 0.2546 kg/m^3
g = 9.81 m/s^2
Taper ratio = 1
b = 30 m
c = 4 m
m = 70,000 kg
c_L = 0.85
V = 229.97 m/s
density_{land}=1.225 kg/m^3
$c_{L,\ land}$= 1.15

4) Exercises

$density_{cruise} = 0.2546 \text{ kg/m}^3$
$g = 9.81 \text{ m/s}^2$
Taper ratio = 1
$b = 30$ m
$c = 4$ m
$m = 70{,}000$ kg
$c_L = 0.85$
$V = 229.97$ m/s
$density_{land} = 1.225 \text{ kg/m}^3$
$c_{L,\ land} = 1.15$
$A = 7.5$
$e = 0.92$
$c_{D0} = 0.00685$

a) At what speed is the aircraft flying at? (m/s)
b) The aircraft now approaches landing at sea level, and reduces its velocity to 90 m/s. What is the lift coefficient during landing?
c) How could the aircraft achieve this increase in c_L
d) What is the total drag of the aircraft when landing

4) Exercises

a) At what speed is the aircraft flying at? (m/s)
b) The aircraft now approaches landing at sea level, and reduces its velocity to 90 m/s. What is the lift coefficient during landing?
c) How could the aircraft achieve this increase in c_L
d) What is the total drag of the aircraft
e) What is the lift to drag ratio?

$density_{cruise} = 0.2546\ kg/m^3$
$g = 9.81\ m/s^2$
Taper ratio = 1
$b = 30\ m$
$c = 4\ m$
$m = 70{,}000\ kg$
$c_L = 0.85$
$V = 229.97\ m/s$
$density_{land} = 1.225\ kg/m^3$
$c_{L,\ land} = 1.15$
$A = 7.5$
$e = 0.92$
$c_{D0} = 0.00685$
$D = 40{,}400\ N$

4) Exercises

a) At what speed is the aircraft flying at? (m/s)
b) The aircraft now approaches landing at sea level, and reduces its velocity to 90 m/s. What is the lift coefficient during landing?
c) How could the aircraft achieve this increase in c_L
d) What is the total drag of the aircraft
e) What is the lift to drag ratio?

$density_{cruise} = 0.2546\ kg/m^3$
$g = 9.81\ m/s^2$
Taper ratio = 1
$b = 30\ m$
$c = 4\ m$
$m = 70{,}000\ kg$
$c_L = 0.85$
$V = 229.97\ m/s$
$density_{land} = 1.225\ kg/m^3$
$c_{L,\ land} = 1.15$
$A = 7.5$
$e = 0.92$
$c_{D0} = 0.00685$
$D = 40{,}400\ N$
$c_L/c_D = 17$

5) Exercises

The increase in drag caused by shock waves on an airfoil is mainly due to pressure drag.

(a) True

(b) False

Streamlining an object generally reduces the pressure drag.

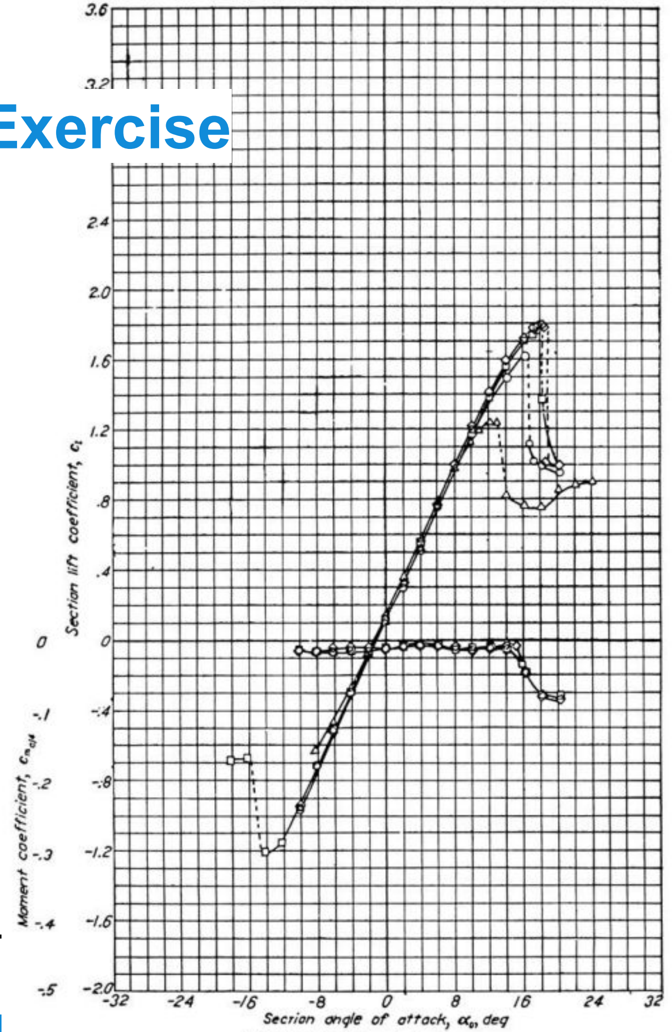
(a) True

(b) False

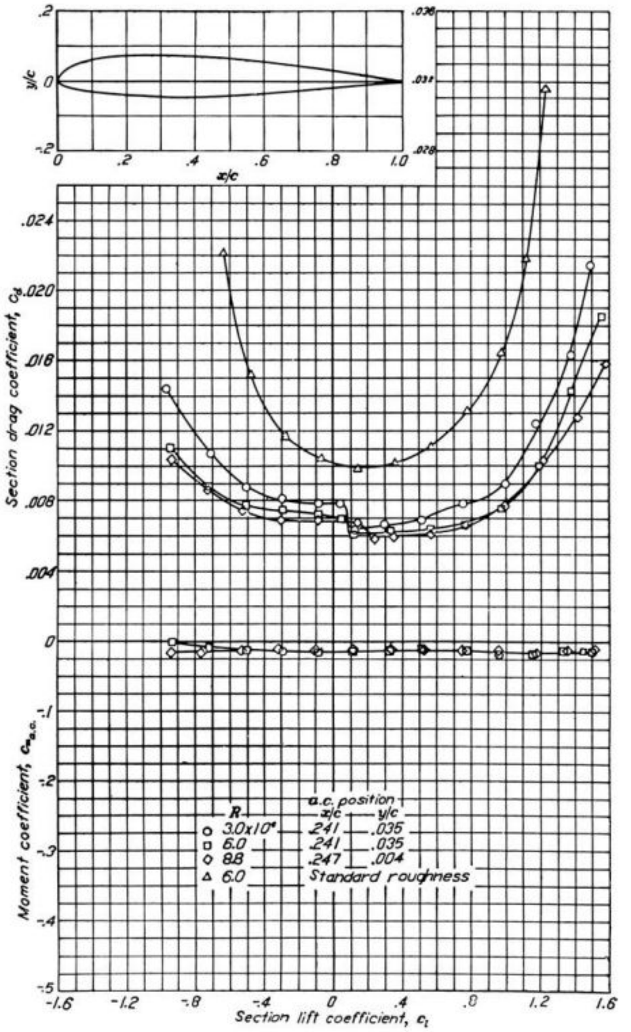
A) Exercises

5.3 Consider a rectangular wing mounted in a low-speed subsonic wing tunnel. The wing model completely spans the test-section so that the flow "sees" essentially an infinite wing. If the wing has a NACA 23012 airfoil section and a chord of 0.3 m, calculate the lift, drag, and about the quarter-chord per unit span when the airflow pressure, temperature, and velocity are 1 atm, 303 K, and 42 m/s, respectively. The angle of attack is 8°.

A) Exercise

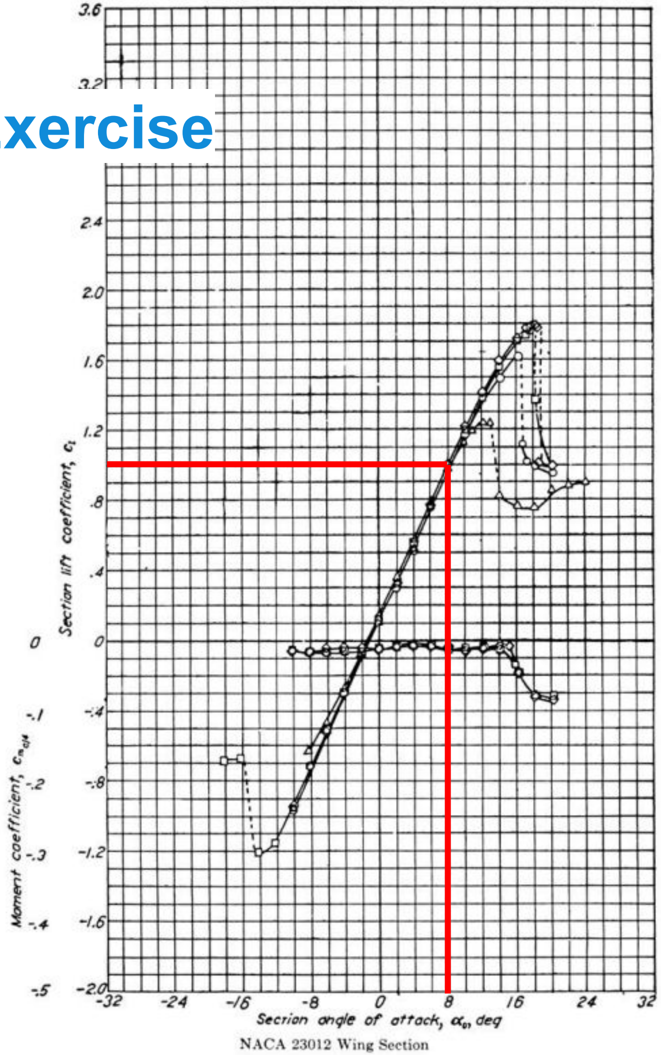


NACA 23012 Wing Section

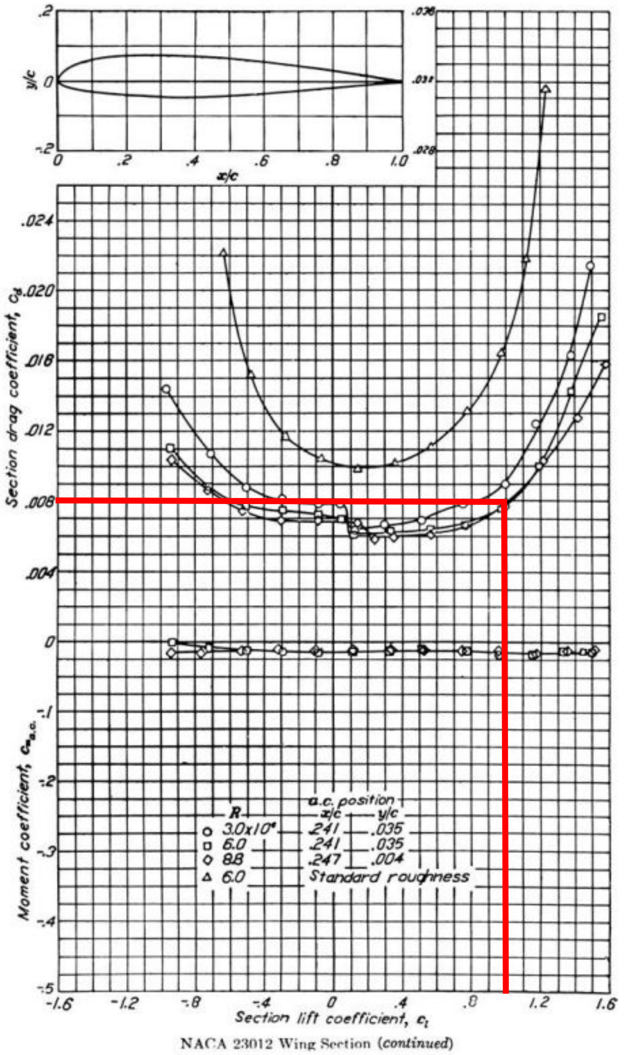


NACA 23012 Wing Section (continued)

1) Exercise



NACA 23012 Wing Section



NACA 23012 Wing Section (continued)